ANOMALY DETECTION IN SECURITY LOGS

Evaluating Detection Models from Simulation to Real-World Cybersecurity Data

Authors
Maheen Syeda, Yi Yang, Ph.D.

Affiliations
Department of Computer Science, NEIU

Problem & Motivation

Security logs are too large and complex to monitor manually. Machine learning can help detect suspicious behavior, but real-world traffic is noisy, constantly changing and difficult to model.

Key Issues

- Huge log volume
- Unknown attack patterns
- High false positives
- Noisy real-world traffic

Security Logs → ML Detection Model → Anomaly Alerts

Too many false positives = analyst overload

Objective

Compare anomaly detection models on simulated security logs and real-world CICIDS2017 traffic.

Focus

- Detection accuracy
- False positives
- Model reliability
- Real-world performance

Methodology

01 Data Collection
Simulated logs + CICIDS2017 network traffic

02 Preprocessing
Missing values removed and labels converted to binary

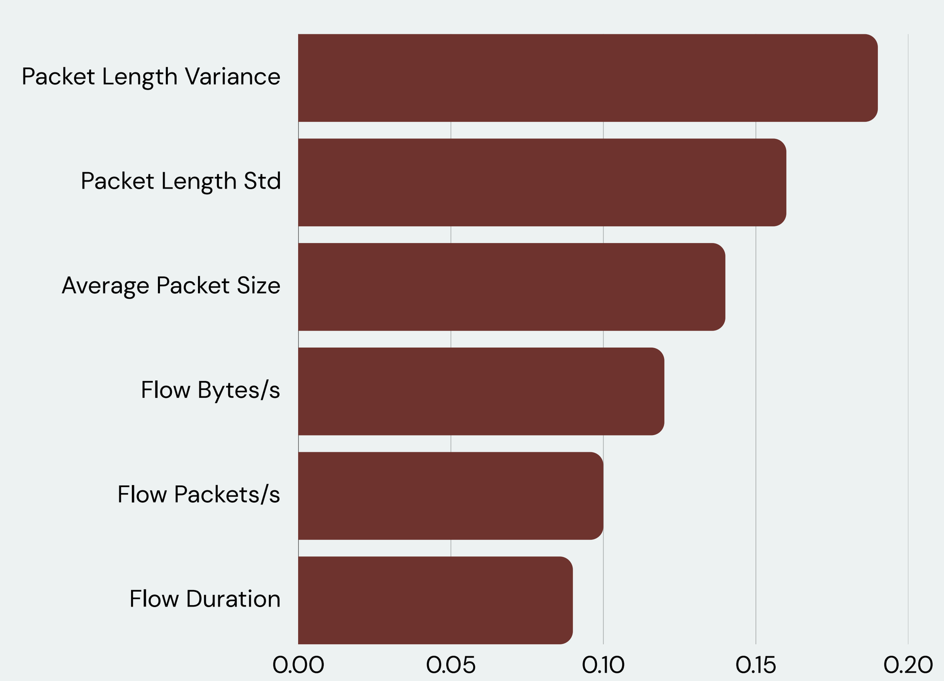
03 Feature Selection
Reduced 53 columns to 13 numerical traffic features

04 Model Training
LOF, One-Class SVM, Isolation Forest and Random Forest

05 Evaluation
Precision, recall, F1 score, ROC curve and dashboard output

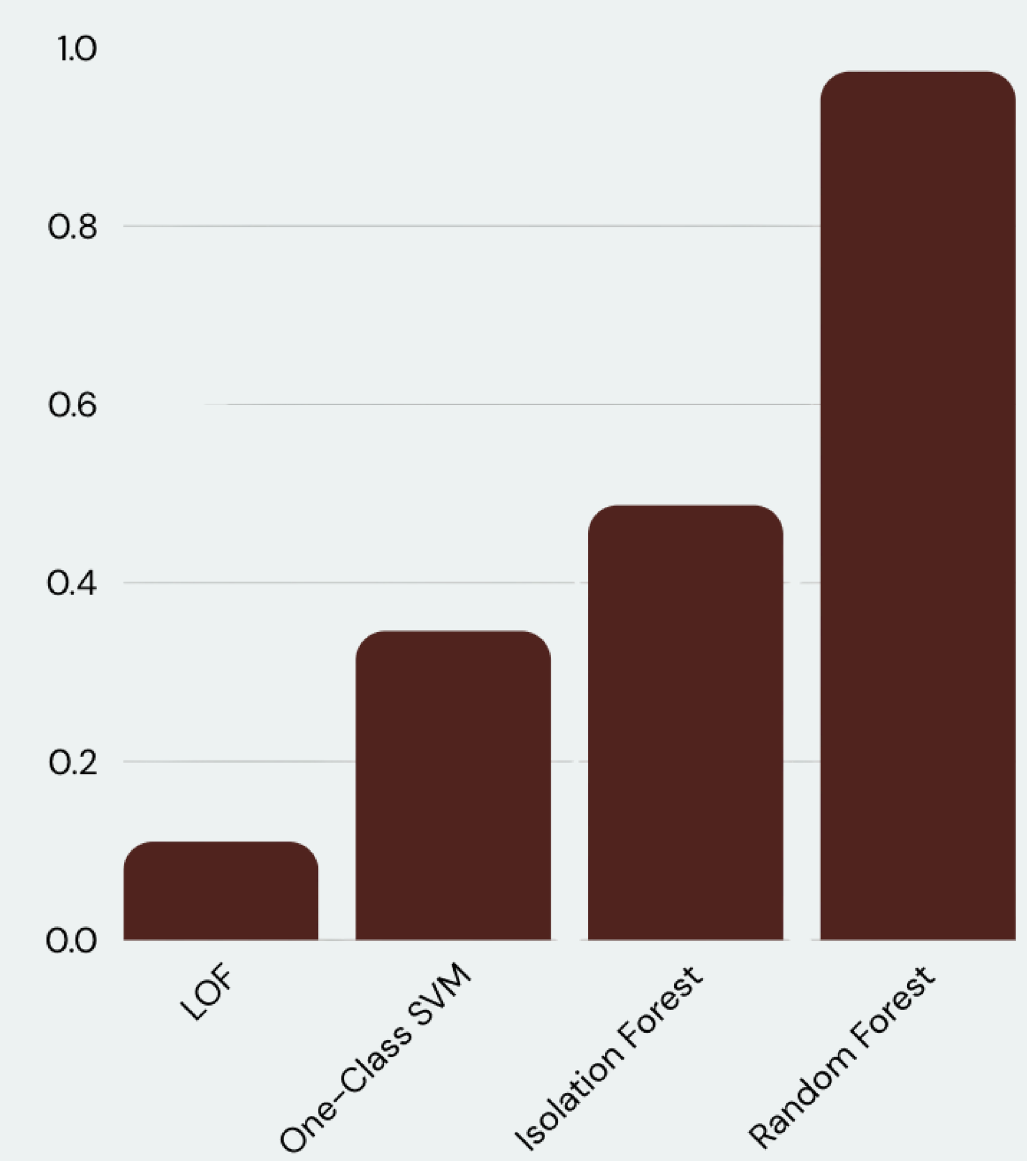
Best Model: Random Forest
F1 Score: 0.973
Accuracy: 0.990

Top Features Used by Random Forest



Precision	Recall	F1 Score	Accuracy
0.970	0.976	0.973	0.990

	Predicted Normal	Predicted Attack
Actual Normal	829	5
Actual Attack	4	162



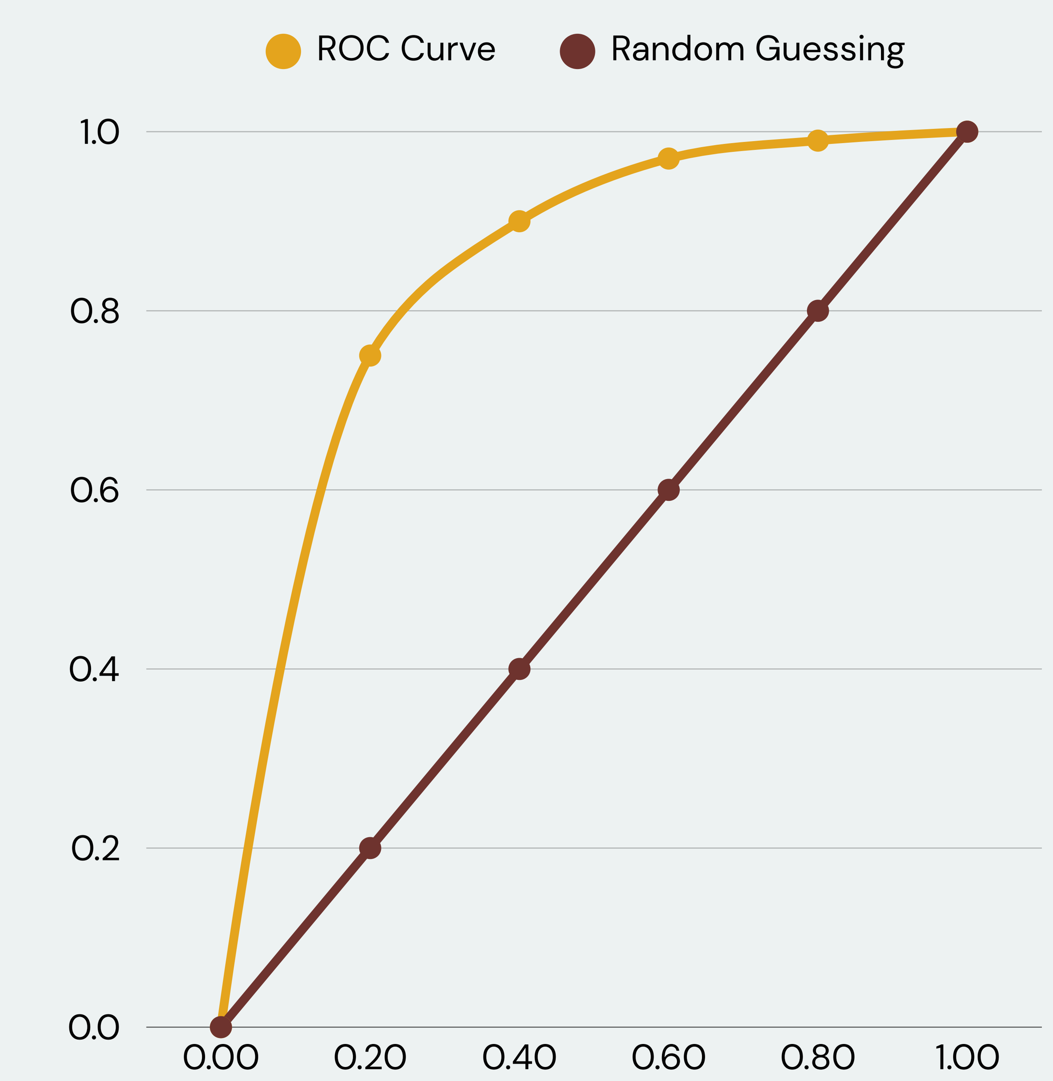
Results

Random Forest achieved the strongest overall detection performance with F1 = 0.973. Isolation Forest performed best among the unsupervised models.

Dataset: CICIDS2017
Original size: 2,520,751 rows × 53 columns
Working sample: 5,000 rows × 53 columns
Selected features:13 numerical traffic features
Labels: 0 = Normal, 1 = Attack

ROC Curve Analysis

Random Forest clearly separated normal traffic from attack traffic.



Key Finding: The model showed strong classification performance across thresholds.

Conclusion

Random Forest provided the strongest detection results. Unsupervised models remain useful when labeled attack data is unavailable.

Key Sources & Acknowledgements

Selected References: Breunig et al. (2000); Liu et al. (2008); Schölkopf et al. (2001); Sharafaldin et al. (2018); Pedregosa et al. (2011)

Acknowledgements: Thank you to Dr. Yi Yang, Dr. Li Wang and the Department of Computer Science at Northeastern Illinois University.

ABET Computing Accreditation Commission

Northeastern Illinois University Computer Science